UNIT-1

1.1 The Internet

The internet is a global network connecting millions of computers.
It provides international communication and services.

International Internet Service Providers

At the top of the hierarchy are international service providers connecting nations.

National Internet Service Providers

Backbone networks created and maintained by specialized companies.

1.2 Data Communication

· Data communications are the exchange of data between two devices via some form of transmission medium.
· The effectiveness of a data communications system depends on four fundamental characteristics:
1. Delivery: The system must deliver data to the correct destination. Data must be received

by the intended device or user and only by that device or user.

2. Accuracy: The system must deliver the data accurately. Data altered in transmission are unusable.

3. Timeliness: The system must deliver data in a timely manner. Late data are useless.

4.

Jitter: Jitter refers to the variation in packet arrival time. It is uneven delay in audio/video delivery.

Components of Data Communication

A data communications system has five components:
1. Message: The information to be communicated.
2. Sender: The device that sends the data.
3. Receiver: The device that receives the message.
4. Transmission medium: Physical path for transmission.
5. Protocol: Set of rules governing data communication.